Saviours AI

ICSE CLASS 10 · 2027 SYLLABUS

SIDE KICK

Chemistry

Nine chapters written fresh for the 2027 syllabus: reactions, equations, observations and HOT questions.



EVERY TOPPER HAS A SIDEKICK

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Hey. I'm your Sidekick.

I live inside this book. Wherever things get tricky, I pop up with a tip, a trap to dodge, or a question the board loves to ask. Read the chapter, then try my HOT questions. Deal?



Aligned to the official ICSE 2027 syllabus. If it's in the syllabus, it's in here.

SIDEKICK TIP
Shortcuts, mnemonics and exam hacks.

FUN FACT
Things that make the concept stick.

REAL LIFE
Where this shows up outside the exam hall.

KEY POINT
Definitions and laws you must write exactly.

HOT QUESTION
Higher-order thinking. Try before you peek.

✦ NEW
Added in this edition to match the 2027 syllabus.

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01

CHAPTER · CHEMISTRY

Periodic Table and Periodic Properties

Elements are not a random crowd. Arrange them by atomic number and their behaviour repeats in a pattern. This chapter shows you the pattern and the reasons behind it.



IN THIS CHAPTER

- Modern periodic law
- Groups and periods
- Six periodic properties
- Alkali metals
- Halogens

The Modern Periodic Law

Mendeleev sorted the elements by atomic mass. The modern table sorts them by **atomic number**, the number of protons, because that number fixes the electron arrangement and so the chemistry.

KEY POINT

Modern periodic law: the physical and chemical properties of elements are a periodic function of their atomic numbers.

- The table has **18 vertical columns called groups** (numbered 1 to 18) and **7 horizontal rows called periods**.
- Elements in one group have the **same number of electrons in the outermost shell**, so they behave alike.
- Across a period the outer shell is the same, but the number of electrons in it grows from 1 to 8.
- The syllabus sets questions only up to period 4, and none on elements beyond calcium. You should still be able to read the whole table.

PERIOD	NUMBER OF ELEMENTS	STARTS WITH	ENDS WITH
1	2	Hydrogen	Helium
2	8	Lithium	Neon
3	8	Sodium	Argon
4	18	Potassium	Krypton

SIDEKICK TIP

The period number tells you how many **shells** an atom has. The group number (for groups 1 and 2, and 13 to 18 as group minus 10) tells you the number of **outer electrons**.

Two Forces That Explain Every Trend

Every periodic trend comes from a tug of war between two things. Keep them in mind and you can rebuild any answer in the exam.

1. Nuclear charge

The positive pull of the nucleus grows as the atomic number grows. A stronger pull draws the electrons in tighter.

2. Number of shells

Every new shell puts the outer electrons farther from the nucleus and shields them from its pull. More distance means a weaker grip.

The Six Properties You Must Know

PROPERTY	WHAT IT MEANS	ACROSS A PERIOD (LEFT TO RIGHT)	DOWN A GROUP
Atomic size	Distance from the nucleus to the outermost electron	**Decreases.** Nuclear charge rises, shells stay the same	**Increases.** A new shell is added each time
Metallic character	Tendency to lose electrons and form positive ions	**Decreases**	**Increases**
Non-metallic character	Tendency to gain electrons and form negative ions	**Increases**	**Decreases**
Ionisation potential	Energy needed to remove the most loosely held electron from a gaseous atom	**Increases.** Electrons are held harder	**Decreases.** Outer electron is farther and shielded
Electron affinity	Energy released when a gaseous atom gains an electron	**Increases** (halogens gain most easily)	**Decreases**
Electronegativity	Power of an atom to attract the shared pair of electrons in a bond	**Increases**	**Decreases**



BOARD ALERT

Do not say ionisation potential 'increases down a group because atoms get bigger'. Bigger atoms release their outer electron more easily, so it **decreases**.

Group 1: The Alkali Metals

Lithium, sodium, potassium, rubidium, caesium and francium. Each has **one electron in its outer shell**, which it loses very readily to form a +1 ion.

- Soft, silvery, low melting point and low density (lithium, sodium and potassium float on water).
- Very reactive. **Reactivity increases down the group** because the single outer electron is lost more easily as the atom grows.
- They react with water to give an alkali and hydrogen: $2Na + 2H_2O \rightarrow 2NaOH + H_2$.
- They are stored under oil because they react with air and moisture.

Group 17: The Halogens

Fluorine, chlorine, bromine, iodine and astatine. Each has **seven electrons in the outer shell** and needs just one more to complete an octet.

- They are diatomic molecules: F_2, Cl_2, Br_2, I_2.

- Colour deepens and the state changes down the group: fluorine is a pale yellow gas, chlorine a greenish-yellow gas, bromine a red-brown liquid and iodine a dark solid.
- **Reactivity decreases down the group**, because the smaller atoms attract an extra electron better.
- A more reactive halogen displaces a less reactive one from its salt: Cl_2 + 2KBr → 2KCl + Br_2.



FUN FACT

Francium is so rare and radioactive that there may be less than a few dozen grams of it in the whole of the Earth's crust at any time.



HOT QUESTION

Element X is in group 1 and period 3, and element Y is in group 17 and period 3. Which has the larger atomic size, which has the higher ionisation potential, and what type of bond forms between them?

02

CHAPTER · CHEMISTRY

Chemical Bonding

Atoms bond to reach the stable electron arrangement of a noble gas. They can give electrons away, share them, or have one atom donate both. Each route builds a different kind of compound.



IN THIS CHAPTER

- Electrovalent bond
- Covalent bond
- Polar and non-polar
- Coordinate bond
- Comparing compounds

Why Atoms Bond

Noble gases have a full outer shell (a duplet of 2 for helium, an octet of 8 for the others) and rarely react. Other atoms tidy up their outer shell by losing, gaining or sharing electrons. That exchange is the chemical bond.

1. Electrovalent (Ionic) Bond

A metal **transfers** its outer electrons to a non-metal. The metal becomes a positive ion (cation), the non-metal a negative ion (anion), and the opposite charges attract strongly.

COMPOUND	ELECTRON TRANSFER	IONS FORMED
Sodium chloride, NaCl	Na (2,8,1) gives 1 electron to Cl (2,8,7)	Na^+ (2,8) and Cl^- (2,8,8)
Magnesium chloride, $MgCl_2$	Mg (2,8,2) gives one electron to each of two Cl atoms	Mg^{2+} and 2 Cl^-
Calcium oxide, CaO	Ca (2,8,8,2) gives 2 electrons to O (2,6)	Ca^{2+} and O^{2-}



SIDEKICK TIP

When you draw the electron-dot structure, show the cation in square brackets with its charge, and show the anion in square brackets with eight dots around it, then write the charge outside.

Properties of electrovalent compounds

- **State:** hard crystalline solids, because ions pack in a rigid lattice.
- **Melting and boiling points:** high, since a lot of energy is needed to break the strong attraction between ions.
- **Conductivity:** do not conduct as solids (ions are fixed). They conduct when **molten or dissolved**, because the ions are free to move. This is the link to electrolysis.
- **Solubility:** usually soluble in water and insoluble in organic solvents.

2. Covalent Bond

Two non-metal atoms **share** pairs of electrons so that each reaches a duplet or an octet. One shared pair is a single bond, two a double bond, three a triple bond.

MOLECULE	BOND	HOW THE OCTET IS REACHED
Hydrogen, H_2	Single	Each H shares 1 pair to get a duplet
Chlorine, Cl_2	Single	Each Cl shares 1 pair; 3 lone pairs remain on each
Oxygen, O_2	Double	Two pairs are shared
Nitrogen, N_2	Triple	Three pairs are shared
Methane, CH_4	4 single	Carbon shares one pair with each of 4 hydrogens
Carbon tetrachloride, CCl_4	4 single	Carbon shares one pair with each of 4 chlorines

Polar and non-polar covalent compounds

If the two atoms are identical, the shared pair sits midway and the molecule is **non-polar** (H_2, Cl_2, O_2, N_2, CH_4, CCl_4). If one atom has a higher electronegativity it pulls the pair closer and the molecule is **polar**, with a slightly negative end and a slightly positive end.

- **HCl:** chlorine is more electronegative, so the pair lies nearer to it.
- **H_2O:** oxygen pulls two pairs towards itself. The molecule is bent.
- **NH_3:** nitrogen pulls three pairs and keeps one lone pair. The molecule is pyramidal.

Properties of covalent compounds

- Gases, liquids or soft solids, since the forces between molecules are weak.
- Low melting and boiling points.
- Do not conduct electricity (no free ions). Polar ones such as HCl, though, ionise in water and then conduct.
- Usually soluble in organic solvents and insoluble in water.

3. Coordinate (Dative) Bond

A coordinate bond is a covalent bond in which **both electrons of the shared pair come from one atom**. The atom that gives is the donor and it must have a **lone pair**. The atom that receives must have an empty space to accept it.

1. Hydronium ion, H_3O^+

A water molecule has two lone pairs on oxygen. One lone pair is donated to a hydrogen ion (H^+), which has no electrons. The result is H_3O^+.

2. Ammonium ion, NH_4^+

Ammonia has one lone pair on nitrogen. It is donated to H^+ and forms NH_4^+.

3. Hydroxide ion, OH^-

When water ionises, a hydrogen ion leaves with no electron and the oxygen keeps the shared pair, which leaves OH^-.

Electrovalent versus Covalent at a Glance

FEATURE	ELECTROVALENT	COVALENT
Formed by	Transfer of electrons	Sharing of electrons
Between	Metal and non-metal	Non-metal and non-metal
State	Crystalline solid	Gas, liquid or soft solid
Melting and boiling points	High	Low
Conducts electricity	Molten or dissolved only	Generally no
Solubility	Soluble in water	Soluble in organic solvents
Reactions	Fast ionic reactions	Slower, between molecules



HOT QUESTION

Carbon tetrachloride does not conduct electricity in any state, but hydrogen chloride gas gives a conducting solution in water. Explain the difference using the type of bond.

03

CHAPTER · CHEMISTRY

Acids, Bases and Salts

Lemon juice, soap and table salt are all members of one family tree. Learn what each of them releases in water, how to test for them and how to make a salt on demand.



IN THIS CHAPTER

- Acids and alkalis
- Indicators and pH
- Types of salts
- Action of acids on salts
- Preparing salts

What Makes an Acid, an Alkali and a Salt

- An **acid** is a substance that gives **hydronium ions (H_3O^+)** as the only positive ions when dissolved in water. $HCl + H_2O \rightarrow H_3O^+ + Cl^-$.
- An **alkali** is a base that is soluble in water and gives **hydroxide ions (OH^-)** as the only negative ions. $NaOH \rightarrow Na^+ + OH^-$.
- A **salt** is formed when the hydrogen of an acid is replaced, partly or completely, by a metal or by the ammonium ion.

	ACID	ALKALI	SALT
Ion in water	Hydronium (positive)	Hydroxide (negative)	Metal ion and acid-radical ion
Example	HCl, H_2SO_4, HNO_3	NaOH, KOH, NH_4OH	NaCl, $CuSO_4$
Effect on litmus	Blue turns red	Red turns blue	No change



KEY POINT

The basicity of an acid is the number of hydrogen ions it can give. HCl and HNO_3 are monobasic, H_2SO_4 is dibasic, and H_3PO_4 is tribasic.

Indicators and the pH Scale

INDICATOR	IN ACID	IN ALKALI
Litmus	Red	Blue
Methyl orange	Red	Yellow
Phenolphthalein	Colourless	Pink
Universal indicator	Red to orange	Blue to violet

The pH scale runs from 0 to 14. **Below 7 is acidic, exactly 7 is neutral, and above 7 is alkaline**. A lower pH means a stronger acid and a higher pH means a stronger alkali. pH paper or universal indicator shows the value as a colour.



SIDEKICK TIP

Phenolphthalein is the odd one out. It is colourless in acid and neutral solutions and only turns pink in an alkali.

Types of Salts

TYPE	HOW IT FORMS	EXAMPLE
Normal salt	All hydrogen of the acid replaced	NaCl, Na_2SO_4, $CuSO_4$
Acid salt	Only part of the hydrogen replaced, so it still has replaceable H	$NaHSO_4$, $NaHCO_3$
Basic salt	Contains the hydroxide or oxide radical along with the acid radical	Basic lead chloride Pb(OH)Cl
Complex salt	Contains a complex ion made with a central metal ion	Sodium zincate Na_2ZnO_2; tetraammine copper(II) sulphate

Dilute Acids Acting on Salts

A dilute acid decomposes a salt of a weaker or more volatile acid. Gently heat if needed.

SALT	GAS RELEASED	TEST FOR THE GAS	EXAMPLE EQUATION
Carbonate	Carbon dioxide	Turns lime water milky	$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$
Hydrogen carbonate	Carbon dioxide	Turns lime water milky	$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$
Sulphite	Sulphur dioxide	Turns acidified potassium dichromate from orange to green; suffocating smell	$Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2$
Hydrogen sulphite	Sulphur dioxide	As above	$NaHSO_3 + HCl \rightarrow NaCl + H_2O + SO_2$
Sulphide	Hydrogen sulphide	Rotten-egg smell; turns lead acetate paper black	$ZnS + H_2SO_4 \rightarrow ZnSO_4 + H_2S$

Six Ways to Prepare a Normal Salt

METHOD	USE IT WHEN	EXAMPLE
Direct combination	Making a salt from its two elements	$Fe + S \rightarrow FeS$ (heat). $2Fe + 3Cl_2 \rightarrow 2FeCl_3$
Displacement	A more reactive metal displaces hydrogen from a dilute acid	$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
Precipitation (double decomposition)	Both starting salts are soluble and the product is insoluble	$AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3$
Neutralisation of an insoluble base	The base does not dissolve, so add it in excess and filter	$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$
Neutralisation of an alkali (titration)	Making sodium, potassium or ammonium salts	$NaOH + HCl \rightarrow NaCl + H_2O$
Acid on a carbonate or bicarbonate	Making a salt and releasing CO_2	$CuCO_3 + H_2SO_4 \rightarrow CuSO_4 + H_2O + CO_2$



BOARD ALERT

An alkali is added drop by drop in a titration, because an excess would spoil the salt. For an insoluble base or carbonate the opposite is true. Add excess solid and filter.



HOT QUESTION

Which method would you choose to make (a) copper sulphate from copper oxide, (b) sodium sulphate, and (c) lead chloride? Write the equation each time.

04

CHAPTER · CHEMISTRY

Analytical Chemistry

Analytical chemistry is detective work. You add a reagent, watch the colour of the precipitate and decide which metal ion is hiding in the solution.



IN THIS CHAPTER

- Colours of salts
- Action of NaOH
- Action of NH4OH
- Special cases
- Amphoteric substances

The Detective Method

1. Look
Note the colour of the salt and of its solution.

2. Add NaOH drop by drop
Observe the colour of the precipitate. Then add excess and see whether it dissolves.

3. Repeat with NH_4OH
Again, observe the precipitate and what happens in excess.

4. Decide
Match your observations with the table below.

ION	COLOUR OF SALT SOLUTION	ADDED NAOH	EXCESS NAOH	ADDED NH_4OH	EXCESS NH_4OH
Copper, Cu^{2+}	Blue	Pale blue precipitate	Insoluble	Pale blue precipitate	**Soluble.** Deep (inky) blue solution
Iron(II), Fe^{2+}	Pale green	Dirty green precipitate	Insoluble	Dirty green precipitate	Insoluble
Iron(III), Fe^{3+}	Yellow or brown	Reddish-brown precipitate	Insoluble	Reddish-brown precipitate	Insoluble
Zinc, Zn^{2+}	Colourless	White precipitate	**Soluble.** Colourless solution	White precipitate	**Soluble.** Colourless solution
Lead, Pb^{2+}	Colourless	White precipitate	**Soluble.** Colourless solution	White precipitate	Insoluble



SIDEKICK TIP
The only metals that dissolve in excess NaOH are **zinc and lead**. Lead fails in NH_4OH while zinc succeeds. That single difference separates them.

Equations You Must Know

With sodium hydroxide

- $CuSO_4 + 2NaOH \rightarrow Cu(OH)_2\downarrow + Na_2SO_4$
- $FeSO_4 + 2NaOH \rightarrow Fe(OH)_2\downarrow + Na_2SO_4$
- $FeCl_3 + 3NaOH \rightarrow Fe(OH)_3\downarrow + 3NaCl$
- $ZnSO_4 + 2NaOH \rightarrow Zn(OH)_2\downarrow + Na_2SO_4$, then $Zn(OH)_2 + 2NaOH \rightarrow Na_2ZnO_2 + 2H_2O$
- $Pb(NO_3)_2 + 2NaOH \rightarrow Pb(OH)_2\downarrow + 2NaNO_3$, then $Pb(OH)_2 + 2NaOH \rightarrow Na_2PbO_2 + 2H_2O$

With ammonium hydroxide

- $CuSO_4 + 2NH_4OH \rightarrow Cu(OH)_2\downarrow + (NH_4)_2SO_4$
- In excess: $Cu(OH)_2 + (NH_4)_2SO_4 + 2NH_4OH \rightarrow [Cu(NH_3)_4]SO_4 + 4H_2O$
- $ZnSO_4 + 2NH_4OH \rightarrow Zn(OH)_2\downarrow + (NH_4)_2SO_4$ (dissolves in excess)
- $FeSO_4$, $FeCl_3$ and $Pb(NO_3)_2$ give their hydroxides, which do not dissolve in excess

Two Special Actions

REACTION	WHAT HAPPENS	WHY IT MATTERS
NH_4OH on a copper salt	A pale blue precipitate forms, then dissolves in excess to give a deep blue solution	Used to confirm copper
NaOH on an ammonium salt	On warming, ammonia gas is released: $NH_4Cl + NaOH \rightarrow NaCl + H_2O + NH_3$	Ammonia has a sharp smell, turns moist red litmus blue and makes dense white fumes near a rod dipped in HCl. Used to confirm the ammonium ion

Amphoteric Substances

An **amphoteric** substance reacts with both an acid and an alkali. Aluminium, zinc and lead, together with their oxides and hydroxides, are amphoteric. With a strong alkali they give a salt in which the metal is part of the negative ion.

SUBSTANCE	WITH CAUSTIC ALKALI	PRODUCT
Zinc	$Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$	Sodium zincate and hydrogen
Aluminium	$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$	Sodium aluminate and hydrogen
Aluminium oxide	$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$	Sodium aluminate
Zinc oxide	$ZnO + 2NaOH \rightarrow Na_2ZnO_2 + H_2O$	Sodium zincate
Lead(II) oxide	$PbO + 2NaOH \rightarrow Na_2PbO_2 + H_2O$	Sodium plumbite
Aluminium hydroxide	$Al(OH)_3 + NaOH \rightarrow NaAlO_2 + 2H_2O$	Sodium aluminate



KEY POINT

Amphoteric = reacts with **both** acids and alkalis. Learn the three metals: aluminium, zinc, lead (Al, Zn, Pb).

HOT QUESTION

A colourless solution gives a white precipitate with NaOH that dissolves in excess. With NH_4OH the white precipitate does not dissolve in excess. Which ion is present, and what is the formula of the soluble product in excess NaOH?

05

CHAPTER · CHEMISTRY

Mole Concept and Stoichiometry

Atoms are far too small to count one by one, so chemists count them in huge bundles called moles. Master one idea, the mole, and every numerical in this chapter becomes a three-step routine.



IN THIS CHAPTER

- The mole
- Gas laws
- Vapour density
- Empirical formula
- Equation calculations

The Mole: Chemistry's Dozen

A dozen means 12 things. A **mole** means **6.022×10^{23}** things, a number named after Avogadro. The things can be atoms, molecules or ions.

TERM	MEANING	EXAMPLE
Atomic mass	Mass of one atom compared with 1/12 of a carbon-12 atom. A pure number	O = 16, C = 12
Gram atomic mass	Mass of **one mole of atoms**, in grams. This amount is also called one gram atom	16 g of oxygen atoms
Molecular mass	Sum of the atomic masses in a molecule. A pure number	CO_2 = 12 + 32 = 44
Gram molecular mass	Mass of **one mole of molecules**, in grams. Also called one gram molecule	44 g of CO_2

Atomic masses to use unless the question gives others: H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, Al = 27, S = 32, Cl = 35.5, K = 39, Ca = 40, Fe = 56, Cu = 63.5, Zn = 65.

The Triangle of Conversions

YOU KNOW	YOU WANT	FORMULA
Mass (g)	Moles	moles = mass / molar mass
Volume of gas at STP (litres)	Moles	moles = volume / 22.4
Number of particles	Moles	moles = number / (6.022×10^{23})



KEY POINT

One mole of **any gas** occupies **22.4 litres at STP** (0 degrees Celsius and 1 atmosphere), and its mass equals its gram molecular mass.

Gas Laws

- **Gay-Lussac's law of combining volumes:** when gases react, they do so in volumes that bear a simple whole-number ratio to one another and to the volumes of gaseous products, all measured at the same temperature and pressure.
- **Avogadro's law:** equal volumes of all gases, under the same conditions of temperature and pressure, contain the **same number of molecules**.

REACTION	VOLUME RATIO	MEANING
$N_2 + 3H_2 \rightarrow 2NH_3$	1 : 3 : 2	1 litre of nitrogen needs 3 litres of hydrogen and gives 2 litres of ammonia
$H_2 + Cl_2 \rightarrow 2HCl$	1 : 1 : 2	1 litre each gives 2 litres of hydrogen chloride
$N_2 + O_2 \rightarrow 2NO$	1 : 1 : 2	1 litre each gives 2 litres of nitric oxide

Because of Avogadro's law, these results show that hydrogen, oxygen, nitrogen and chlorine are **diatomic**. One molecule of each contains two atoms.

Vapour Density

Vapour density of a gas = mass of a certain volume of the gas / mass of the same volume of hydrogen. It follows that **molecular mass = 2 x vapour density**.

Example: if the vapour density of a gas is 22, its molecular mass is 44.

Empirical and Molecular Formula

- The **empirical formula** shows the simplest whole-number ratio of atoms in a compound.
- The **molecular formula** shows the actual number of atoms. It is a whole-number multiple of the empirical formula: molecular formula = (empirical formula) x n, where n = molecular mass / empirical formula mass.

1. Write percentages as grams
Take 100 g of the compound, so each percentage is a mass.

2. Divide by atomic mass
This gives the moles of each element.

3. Divide by the smallest
The results give the simplest ratio. If any value ends in .5, double all of them.

4. Write the formula
Use the ratio as subscripts. That is the empirical formula.

5. Find n and scale up
Use n = molecular mass / empirical formula mass.

Worked example. A compound has C 40%, H 6.7%, O 53.3% and a molecular mass of 180. Moles: C = 40/12 = 3.33, H = 6.7/1 = 6.7, O = 53.3/16 = 3.33. Dividing by 3.33 gives C : H : O = 1 : 2 : 1, so the empirical formula is **CH_2O** (mass 30). Then n = 180 / 30 = 6, so the molecular formula is **$C_6H_{12}O_6$**.

Worked Numericals

QUESTION	WORKING	ANSWER
Mass of 2.5 mol of CO_2	2.5 × 44	110 g
Volume at STP of 11 g of CO_2	11 / 44 = 0.25 mol; 0.25 × 22.4	5.6 litres
Molecules in 9 g of water	9 / 18 = 0.5 mol; 0.5 × 6.022 × 10^{23}	3.011 × 10^{23}
Moles in 5.6 litres of NH_3 at STP	5.6 / 22.4	0.25 mol

Calculations Based on Equations

A balanced equation is a recipe in moles. For $CaCO_3 \rightarrow CaO + CO_2$: 1 mole (100 g) of calcium carbonate gives 1 mole (56 g) of calcium oxide and 1 mole (22.4 litres at STP) of carbon dioxide.

Example. What mass of CaO and what volume of CO_2 at STP form from 25 g of $CaCO_3$? Moles of $CaCO_3$ = 25/100 = 0.25. So CaO = 0.25 × 56 = **14 g**, and CO_2 = 0.25 × 22.4 = **5.6 litres**.

SIDEKICK TIP

Always balance the equation first. Every mistake in a stoichiometry question starts with an unbalanced equation.

HOT QUESTION

Find the number of moles and the number of molecules in 4.4 g of carbon dioxide, and the volume this gas occupies at STP.

06

CHAPTER · CHEMISTRY

Electrolysis

Electricity can force a chemical change that would not happen by itself. Follow the ions to the electrodes and you can predict every product.



IN THIS CHAPTER

- Electrolytes
- Key terms
- Electrolysis cells
- Selective discharge
- Electroplating and refining

Electrolytes and Non-electrolytes

TYPE	WHAT IT IS	EXAMPLES
Strong electrolyte	Almost completely ionised in solution or in the molten state, so it conducts well	NaCl, HCl, H_2SO_4, NaOH
Weak electrolyte	Only partly ionised, so it conducts poorly. Both molecules and ions are present	Acetic acid, NH_4OH, H_2CO_3
Non-electrolyte	Contains only molecules, no ions, and does not conduct	Sugar, alcohol, glucose, distilled water



SIDEKICK TIP

Think of it in terms of what is in the beaker. **Only ions** (ionic solid): conducts when molten. **Only molecules**: does not conduct. **Molecules and ions**: weak electrolyte.

Words You Need

TERM	MEANING
Electrolysis	Chemical decomposition of an electrolyte by passing electricity through it
Electrode	A rod or plate that carries current into or out of the electrolyte
Anode	The positive electrode, joined to the positive terminal of the battery
Cathode	The negative electrode, joined to the negative terminal
Anion / cation	Negative ion, which moves to the anode / positive ion, which moves to the cathode
Oxidation	Loss of electrons (happens at the anode)
Reduction	Gain of electrons (happens at the cathode)



KEY POINT

Memory line: **AN OX, RED CAT**. Anode = oxidation, reduction = cathode.

Selective Discharge: Who Goes First

When several ions reach an electrode, only one type is discharged at a time. Three factors decide the winner.

- **Position in the electrochemical series.** The ion lower in the series (a less reactive metal, for example Cu^{2+} compared with H^+) is discharged first at the cathode. At the anode, OH^- is discharged ahead of sulphate and nitrate ions.
- **Concentration.** A very concentrated solution of an ion can overcome a poor position in the series.

- **Nature of the electrode.** An inert electrode (platinum or graphite) only carries current. An active electrode (such as copper) may itself dissolve.

The Three Cells in the Syllabus

ELECTROLYTE AND ELECTRODES	IONS PRESENT	AT THE CATHODE (REDUCTION)	AT THE ANODE (OXIDATION)	OBSERVATION
Molten lead bromide, graphite electrodes	Pb^{2+}, Br^-	$Pb^{2+} + 2e^- \rightarrow Pb$	$2Br^- \rightarrow Br_2 + 2e^-$	Grey molten lead at the cathode and reddish-brown bromine fumes at the anode
Acidified water, platinum electrodes	H^+, OH^- (from water and the acid)	$2H^+ + 2e^- \rightarrow H_2$	$4OH^- \rightarrow 2H_2O + O_2 + 4e^-$	Hydrogen and oxygen collect in a volume ratio of 2 : 1
Aqueous copper sulphate, platinum electrodes	Cu^{2+}, H^+, SO_4^{2-}, OH^-	$Cu^{2+} + 2e^- \rightarrow Cu$	$4OH^- \rightarrow 2H_2O + O_2 + 4e^-$	Brown copper coats the cathode, oxygen bubbles at the anode and the blue colour fades as the solution turns acidic
Aqueous copper sulphate, copper electrodes	Cu^{2+}, H^+, SO_4^{2-}, OH^-	$Cu^{2+} + 2e^- \rightarrow Cu$	$Cu \rightarrow Cu^{2+} + 2e^-$ (the anode dissolves)	The anode thins, the cathode thickens and the blue colour stays unchanged



BOARD ALERT

In the copper-electrode cell, no gas is evolved and the colour does not change. That is the key difference from the platinum-electrode cell. Mention it whenever the two are compared.

Applications of Electrolysis

Electroplating

Coating a metal object with a thin layer of another metal, either to prevent corrosion or to improve its look. The **object is made the cathode** and the **plating metal is made the anode**.

PLATING	CATHODE	ANODE	ELECTROLYTE
Nickel plating	Article to be plated	Pure nickel	Nickel sulphate solution
Silver plating	Article to be plated	Pure silver	Sodium argentocyanide, $Na[Ag(CN)_2]$

- Reaction at the cathode: $Ni^{2+} + 2e^- \rightarrow Ni$ (or $Ag^+ + e^- \rightarrow Ag$).
- Reaction at the anode: $Ni \rightarrow Ni^{2+} + 2e^-$ (or $Ag \rightarrow Ag^+ + e^-$).
- A thin, smooth and even coating forms only when the current is low and steady. Silver is plated from a complex cyanide because it releases silver ions slowly.

Electrorefining of copper

Impure copper is made the **anode**, a strip of pure copper the **cathode**, and the electrolyte is copper sulphate acidified with sulphuric acid. Copper dissolves from the anode and is deposited as pure copper on the cathode. The impurities settle below the anode as anode mud.

- Anode: $Cu \rightarrow Cu^{2+} + 2e^-$
- Cathode: $Cu^{2+} + 2e^- \rightarrow Cu$



HOT QUESTION

Why does the mass of the anode decrease and the mass of the cathode increase by the same amount during copper refining, while the concentration of the copper sulphate solution stays nearly constant?

07

CHAPTER · CHEMISTRY

Metallurgy

Metals do not come out of the ground ready to use. Metallurgy is the step-by-step journey from a rock to a pure metal. Learn the stages, and the extra twists for aluminium.



IN THIS CHAPTER

- Ores and minerals
- Concentration
- Roasting and calcination
- Reduction
- Aluminium
- Alloys

Minerals and Ores

- A **mineral** is any natural substance in which a metal or its compound is found in the earth.
- An **ore** is a mineral from which the metal can be extracted **profitably**. Every ore is a mineral, but not every mineral is an ore.
- **Gangue** (or matrix) is the sand, clay and rocky waste mixed with the ore.

METAL	COMMON ORES	FORMULA
Iron	Haematite; Magnetite	Fe_2O_3; Fe_3O_4
Aluminium	Bauxite	$Al_2O_3.2H_2O$
Zinc	Zinc blende; Calamine	ZnS; $ZnCO_3$

Stages of Extraction

1. 1. Dressing the ore
Crush the ore and remove the gangue to concentrate it.

2. 2. Conversion to oxide
Roast or calcine the concentrated ore to change it into its oxide.

3. 3. Reduction
Remove the oxygen from the oxide to leave the free metal.

4. 4. Refining
Purify the metal, for example by electrorefining.

Step 1: Ways to concentrate the ore

METHOD	PRINCIPLE	USED FOR
Hydraulic washing (gravity separation)	A stream of water washes away the lighter gangue and leaves the heavier ore behind	Oxide ores such as haematite
Magnetic separation	A magnet attracts a magnetic ore and leaves the non-magnetic gangue	Magnetite (iron ore)
Froth flotation	The sulphide ore sticks to oil and rises in a froth with air bubbles, while the gangue sinks in water	Sulphide ores such as zinc blende

Step 2: Roasting and calcination

PROCESS	DEFINITION	EXAMPLE
Roasting	Heating the ore strongly in a **free supply of air** below its melting point. Used for sulphide ores	$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$
Calcination	Heating the ore strongly in a **limited supply or absence of air**. Used for carbonate and hydrated ores, which lose CO_2 or water	$ZnCO_3 \rightarrow ZnO + CO_2$

Step 3: Reduction of the oxide

- **Hydrogen, carbon or carbon monoxide** reduce the oxides of less reactive metals: $CuO + H_2 \rightarrow Cu + H_2O$; $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$; $ZnO + C \rightarrow Zn + CO$; $PbO + C \rightarrow Pb + CO$.
- Very reactive metals hold on to oxygen so tightly that carbon cannot reduce their oxides: **Al_2O_3 and MgO**.
- Those **very active metals** (sodium, potassium, calcium, aluminium) are extracted by the **electrolysis of their molten compounds**.



KEY POINT

The more reactive the metal, the harder it is to extract. This is why the very reactive metals were discovered much later than copper and iron.

Extraction of Aluminium

A. Purifying bauxite: Baeyer's process

1. Dissolve

Heat crushed bauxite with concentrated sodium hydroxide. The alumina dissolves, but the iron oxide and silica impurities stay solid. $Al_2O_3.2H_2O + 2NaOH \rightarrow 2NaAlO_2 + 3H_2O$.

2. Filter

Remove the insoluble red mud.

3. Precipitate

Dilute the filtrate and add a little freshly made $Al(OH)_3$ as seed. $NaAlO_2 + 2H_2O \rightarrow Al(OH)_3 + NaOH$.

4. Heat

Strongly heat the precipitate to get pure alumina. $2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$.

B. Electrolytic extraction: Hall-Heroult process

PART OF THE CELL	MATERIAL	PURPOSE
Electrolyte	Pure alumina dissolved in molten **cryolite** (Na_3AlF_6), with some fluorspar (CaF_2)	Cryolite lowers the melting point (from about 2000 to about 950 degrees Celsius) and makes the melt conduct better
Cathode	Graphite lining of the steel tank	Aluminium ions are reduced here
Anode	Graphite rods	Oxide ions are oxidised here

- At the cathode: $Al^{3+} + 3e^- \rightarrow Al$. Molten aluminium sinks to the bottom and is tapped off.
- At the anode: oxygen is released and attacks the hot graphite anodes, forming CO and CO_2. The anodes burn away slowly and must be replaced.



SIDEKICK TIP

Memory line for the cell: **alumina + cryolite + carbon**. Cryolite reduces the temperature, carbon is both electrodes.

Alloys and Amalgams

An **alloy** is a homogeneous mixture of a metal with one or more other metals or non-metals. An **amalgam** is an alloy in which one metal is **mercury**.

ALLOY	COMPOSITION	USES
Stainless steel	Iron with chromium, nickel and carbon	Utensils, surgical instruments, cutlery
Magnalium	Aluminium with 5% magnesium	Aircraft parts, light scientific apparatus
Duralumin	Aluminium with copper, magnesium and manganese	Aircraft and vehicle bodies
Brass	Copper and zinc	Utensils, electrical fittings, musical instruments
Bronze	Copper and tin	Statues, medals, bells, coins
Fuse metal / solder	Lead and tin (fuse metal also uses bismuth)	Fuse wires, joining metals



HOT QUESTION

A student says that cryolite is added during the electrolysis of alumina because it is a reducing agent. Why is this wrong, and what is its true role?

08

CHAPTER · CHEMISTRY

Study of Compounds

Four industrial giants: hydrogen chloride, ammonia, nitric acid and sulphuric acid. For each one, learn how it is made, how it behaves and how to prove it is there.



IN THIS CHAPTER

- Hydrogen chloride
- Ammonia
- Nitric acid
- Sulphuric acid

The syllabus asks you to study every reaction in terms of **reactants, products, conditions, observations and a balanced equation**, with diagrams where relevant.

A. Hydrogen Chloride (HCl)

Laboratory preparation

- **Reactants:** sodium chloride and concentrated sulphuric acid.
- **Equation (below 200 degrees Celsius):** $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$. At a higher temperature $NaHSO_4 + NaCl \rightarrow Na_2SO_4 + HCl$.
- **Drying:** pass through concentrated sulphuric acid.
- **Collection:** by upward delivery (downward displacement of air), because the gas is heavier than air. Never over water, since it is extremely soluble.

Properties and experiments

EXPERIMENT	OBSERVATION	INFERENCE
Fountain experiment (inverted flask of HCl, dip the jet into water containing blue litmus)	Water rushes up as a fountain and the litmus turns red	HCl is extremely soluble in water and its solution is acidic
Density test	The gas pushes air upward and stays low	HCl is heavier than air
Bring a rod dipped in ammonia near the gas	Dense white fumes of NH_4Cl: $NH_3 + HCl \rightarrow NH_4Cl$	Used as a test for both gases

Hydrochloric acid is made by dissolving the gas in water through an **inverted funnel** that only just touches the surface. If the water starts to rise, the funnel mouth leaves the water, the pressure inside balances and the **back suction** is avoided.

- Reacts with metals ($Zn + 2HCl \rightarrow ZnCl_2 + H_2$), oxides, hydroxides and carbonates to give chlorides.
- Decomposes carbonates, hydrogen carbonates, sulphites and sulphides to give CO_2, SO_2 or H_2S.
- **Aqua regia:** 3 parts concentrated HCl to 1 part concentrated HNO_3. It dissolves gold and platinum, which neither acid can dissolve alone.

TEST	WITH	RESULT
HCl acid / chloride solution	Silver nitrate solution	White curdy precipitate of AgCl, soluble in ammonium hydroxide: $HCl + AgNO_3 \rightarrow AgCl\downarrow + HNO_3$
HCl acid / chloride solution	Lead nitrate solution	White precipitate of $PbCl_2$, soluble on heating: $2HCl + Pb(NO_3)_2 \rightarrow PbCl_2\downarrow + 2HNO_3$
HCl gas	Ammonia	Dense white fumes
HCl gas	Moist blue litmus	Turns red

B. Ammonia (NH_3)

Laboratory preparation

- **Reactants:** ammonium chloride and slaked lime (calcium hydroxide), heated.
- **Equation:** $2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3$.
- **Drying:** pass through **quicklime (CaO)**. Never use concentrated H_2SO_4, P_2O_5 or $CaCl_2$, since they react with ammonia.
- **Collection:** by downward delivery (upward displacement of air) because the gas is lighter than air.
- Other sources: $Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$; $AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3$; and any ammonium salt warmed with an alkali.

Manufacture: Haber's process

$N_2 + 3H_2 \rightleftharpoons 2NH_3$ + heat. Conditions: about **200 atmospheres**, about **450 to 500 degrees Celsius**, **finely divided iron** as catalyst with molybdenum as promoter. The ammonia is liquefied and removed, and unreacted gases are recycled.

Properties

REACTION	EQUATION AND OBSERVATION
Fountain experiment	Very soluble in water; the solution turns red litmus blue and phenolphthalein pink
With HCl	$NH_3 + HCl \rightarrow NH_4Cl$ (dense white fumes)
Burning in oxygen	$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$ (yellowish flame)
Catalytic oxidation	$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ (platinum, 800 degrees Celsius)
With hot copper(II) oxide	$2NH_3 + 3CuO \rightarrow 3Cu + N_2 + 3H_2O$ (black turns reddish brown)
With chlorine	Excess ammonia: $8NH_3 + 3Cl_2 \rightarrow 6NH_4Cl + N_2$. Excess chlorine: $NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$
Aqueous ammonia with acids	With H_2SO_4: $2NH_4OH + H_2SO_4 \rightarrow (NH_4)_2SO_4 + 2H_2O$
Aqueous ammonia with metal salts	Gives precipitates of the hydroxides of iron, lead and zinc, and the copper complex as in Analytical Chemistry

Uses: fertilisers (urea, ammonium sulphate), nitric acid, explosives, cleaning agents and refrigeration. Chlorofluorocarbons (CFCs) deplete the ozone layer and are replaced by ozone-friendly alternatives such as HFCs and hydrocarbons. **Test:** pungent smell; turns moist red litmus blue; makes dense white fumes with HCl.

C. Nitric Acid (HNO_3)

- **Laboratory preparation:** heat potassium (or sodium) nitrate with concentrated sulphuric acid in an all-glass retort. $KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$. Keep the temperature low, because strong heating decomposes

the acid, which turns yellow through dissolved NO_2.

- **Manufacture: Ostwald process.** (1) $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ over platinum gauze at about 800 degrees Celsius. (2) $2NO + O_2 \rightarrow 2NO_2$. (3) $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$.

PROPERTY	REACTION
Oxidising agent with copper	Concentrated: $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$ (brown fumes). Dilute: $3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$
Oxidising agent with non-metals	$C + 4HNO_3 \rightarrow CO_2 + 4NO_2 + 2H_2O$; $S + 6HNO_3 \rightarrow H_2SO_4 + 6NO_2 + 2H_2O$
Acidic properties	$Mg + 2HNO_3$ (very dilute) $\rightarrow Mg(NO_3)_2 + H_2$. Reacts with oxides, hydroxides and carbonates: $CuO + 2HNO_3 \rightarrow Cu(NO_3)_2 + H_2O$

Test (brown ring test): add iron(II) sulphate to the solution, then pour concentrated sulphuric acid slowly down the side of the tube. A **brown ring** forms where the layers meet.

D. Sulphuric Acid (H2SO4)

Manufacture: Contact process

1. Make SO_2
Burn sulphur in air: $S + O_2 \rightarrow SO_2$.

2. Oxidise to SO_3
$2SO_2 + O_2 \rightarrow 2SO_3$ + heat, using **vanadium pentoxide (V_2O_5)** at about 450 degrees Celsius and 1 to 2 atmospheres.

3. Absorb in acid
Dissolve SO_3 in concentrated H_2SO_4 to get oleum: $SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$.

4. Dilute
Add water to the oleum: $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$. ($SO_3$ is not dissolved directly in water because it forms a fine acid mist.)

BEHAVIOUR	DETAILS
Dilute acid	Reacts with metals ($Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$), oxides, hydroxides, carbonates, hydrogen carbonates, sulphites and sulphides. It is **dibasic**, so it forms normal and acid salts (Na_2SO_4 and $NaHSO_4$)
Oxidising agent (concentrated)	$C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$; $S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$
Dehydrating agent	Sugar turns into a black mass of carbon: $C_{12}H_{22}O_{11} \rightarrow 12C + 11H_2O$. Blue copper sulphate crystals turn white: $CuSO_4.5H_2O \rightarrow CuSO_4 + 5H_2O$
Non-volatile	Used to make volatile acids: $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$; $KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$

Test: add barium chloride solution. A **white precipitate of barium sulphate** that is insoluble in dilute hydrochloric acid confirms a sulphate.

BOARD ALERT

When diluting sulphuric acid, always **add the acid slowly to water**, never water to acid, because the reaction gives out a lot of heat.

HOT QUESTION

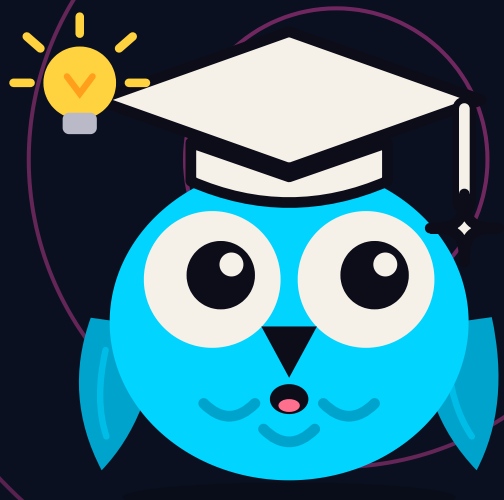
Why is concentrated sulphuric acid used to dry HCl gas but not ammonia? And why is quicklime used to dry ammonia?

09

CHAPTER · CHEMISTRY

Organic Chemistry

Carbon forms more compounds than all other elements combined. Learn why carbon is special, how to name its compounds, and the handful of reactions the syllabus asks for.



IN THIS CHAPTER

- Carbon's special nature
- Isomerism
- Naming
- Alkanes, alkenes, alkynes
- Ethanol
- Acetic acid

Why Carbon Is Special

- **Tetravalency:** carbon has 4 outer electrons and forms 4 covalent bonds.
- **Catenation:** carbon atoms join to each other to make long chains, branched chains and rings.
- Carbon makes **single, double and triple bonds**.
- Chains can be **straight, branched or cyclic**. The only ring you need is **benzene (C_6H_6)**, a six-carbon ring.

Hydrocarbons and Structural Formulae (up to 5 Carbons)

CARBONS	ALKANE (C_NH_{2N+2})	ALKENE (C_NH_{2N})	ALKYNE (C_NH_{2N-2})
1	Methane CH_4	-	-
2	Ethane C_2H_6	Ethene C_2H_4	Ethyne C_2H_2
3	Propane C_3H_8	Propene C_3H_6	Propyne C_3H_4
4	Butane C_4H_{10}	Butene C_4H_8	Butyne C_4H_6
5	Pentane C_5H_{12}	Pentene C_5H_{10}	Pentyne C_5H_8



SIDEKICK TIP

Alkanes: **sa**turated, single bonds only. Alkenes: **d**ouble bond. Alkynes: **t**riple bond. The ending (-ane, -ene, -yne) tells you the bond.

Isomerism

Isomers are compounds with the same molecular formula but different structural formulae.

TYPE	WHAT DIFFERS	EXAMPLE
Chain isomerism	Arrangement of the carbon chain	Butane ($CH_3CH_2CH_2CH_3$) and 2-methylpropane ($CH_3CH(CH_3)CH_3$), both C_4H_{10}
Position isomerism	Position of the double bond, triple bond or functional group	But-1-ene ($CH_2{=}CHCH_2CH_3$) and but-2-ene ($CH_3CH{=}CHCH_3$), both C_4H_8

Pentane (C_5H_{12}) has **three** chain isomers: pentane, 2-methylbutane and 2,2-dimethylpropane.

Homologous Series

A **homologous series** is a family of organic compounds with the same functional group and similar chemical properties, where each member differs from the next by a **CH_2** group (14 mass units).

- All members fit one general formula.
- Physical properties change gradually: boiling and melting points rise with molecular mass.
- Chemical properties are alike because the functional group is the same.
- They can be prepared by similar methods.

Naming Organic Compounds (IUPAC)

1. Find the longest chain
This gives the root: meth, eth, prop, but, pent.

2. Number the carbons
Start from the end that gives the **smallest number** to the functional group or the multiple bond.

3. Add the suffix
The family decides it, as in the table.

4. Write substituents first
For example, 2-methylbutane, with the position number before the name.

FAMILY	FUNCTIONAL GROUP	SUFFIX / PREFIX	EXAMPLE
Halide	-Cl, -Br, -I	chloro, bromo, iodo	Bromoethane C_2H_5Br
Alkene	C=C	-ene	Propene
Alkyne	C≡C	-yne	Ethyne
Alcohol	-OH	-ol	Ethanol C_2H_5OH
Aldehyde	-CHO	-al	Ethanal CH_3CHO
Ketone	C=O (inside the chain)	-one	Propanone CH_3COCH_3
Carboxylic acid	-COOH	-oic acid	Ethanoic acid CH_3COOH
Ether	-O-	alkoxy...alkane	Methoxymethane CH_3OCH_3
Ester	-COO-	-oate	Ethyl ethanoate $CH_3COOC_2H_5$

Alkanes: Methane and Ethane

TOPIC	REACTION
Methane from sodium ethanoate	$CH_3COONa + NaOH \rightarrow CH_4 + Na_2CO_3$ (heat with soda lime)
Ethane from sodium propanoate	$CH_3CH_2COONa + NaOH \rightarrow C_2H_6 + Na_2CO_3$ (soda lime, heat)
From haloalkanes (reduction)	$CH_3I + 2[H] \rightarrow CH_4 + HI$; $C_2H_5Br + 2[H] \rightarrow C_2H_6 + HBr$
Complete combustion	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$; $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$
Incomplete combustion	$2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$ (limited oxygen; poisonous carbon monoxide)
Substitution with chlorine	$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (in sunlight); $C_2H_6 + Cl_2 \rightarrow C_2H_5Cl + HCl$



FUN FACT

Methane is a greenhouse gas, and it is the main part of natural gas and biogas.

Alkenes: Ethene

- **Dehydrohalogenation:** $C_2H_5Br + KOH$ (alcoholic) $\rightarrow C_2H_4 + KBr + H_2O$.
- **Dehydration of ethanol:** $C_2H_5OH \rightarrow C_2H_4 + H_2O$ with concentrated H_2SO_4 at 170 degrees Celsius.
- **Addition reactions:** unsaturated compounds add atoms across the double bond. $C_2H_4 + H_2 \rightarrow C_2H_6$ (nickel catalyst); $C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$ (reddish-brown bromine decolourises, the test for unsaturation).

Alkynes: Ethyne

- **From calcium carbide:** $CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$.
- **From 1,2-dibromoethane:** $CH_2Br\text{-}CH_2Br + 2KOH$ (alcoholic) $\rightarrow C_2H_2 + 2KBr + 2H_2O$.
- **Addition:** $C_2H_2 + H_2 \rightarrow C_2H_4 \rightarrow C_2H_6$, and $C_2H_2 + 2Br_2 \rightarrow C_2H_2Br_4$.

COMPOUND	MAIN USES
Methane	Fuel (CNG, biogas), making hydrogen and carbon black
Ethane	Fuel and making ethene
Ethene	Making polythene, ethanol and ripening fruit
Ethyne	Oxy-acetylene torch for welding and cutting, and making organic compounds

Ethanol

- **Preparation (from alkyl halide):** C_2H_5Br + NaOH (aqueous) → C_2H_5OH + NaBr.
- **Physical:** colourless liquid with a pleasant smell, soluble in water in all proportions, lighter than water, boils at about 78 degrees Celsius.
- **Combustion:** $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$.
- **With sodium:** $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$.
- **Esterification:** $C_2H_5OH + CH_3COOH \rightarrow CH_3COOC_2H_5 + H_2O$ (concentrated H_2SO_4 catalyst).
- **Dehydration:** gives ethene with concentrated H_2SO_4.

Denatured (methylated) spirit is ethanol made undrinkable by adding methanol and a dye, so it cannot be misused. **Spurious alcohol** contains methanol, which causes blindness and can be fatal. Ethanol is used as a solvent, in medicines and perfumes, as a fuel additive and in making other chemicals.

Acetic Acid (Ethanoic Acid)

- **Structure:** CH_3-COOH.
- **Physical:** colourless liquid with the smell of vinegar. Pure acid freezes into ice-like crystals at about 17 degrees Celsius and is called **glacial acetic acid**.
- **Chemical:** turns blue litmus red; reacts with alkalis ($CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$) and with alcohols to form esters (esterification).
- **Uses:** vinegar for food preservation, making dyes and plastics, and as a solvent.

KEY POINT

Esterification is the reaction between an alcohol and a carboxylic acid that gives an ester and water. Esters have a sweet, fruity smell.

HOT QUESTION

How would you distinguish ethane from ethene in the laboratory, and what is the product of the reaction that proves your answer?

Done reading? Now go break the board.



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